

Biological Large Language Models: A Comparative Review of Protein and DNA Foundation Models for AI-Driven Scientific Discovery

Introduction

The adaptation of transformer-based architectures to bioinformatics treats protein and DNA sequences as biological languages (1). Amino acids and nucleic acids act as semantic "words" that construct structural and regulatory "sentences," allowing Natural Language Processing (NLP) to decode the complex grammar of life (2).

However, existing literature reviews discuss protein and genomic models in isolation, omitting rigorous cross-molecular comparison (Proteins vs. DNA). Significant research gaps remain regarding how tokenization strategies influence long-range dependency capture, and how the practical boundaries of Zero-shot Learning (ZSL) operate in data-scarce scenarios.

Key Takeaways

This comparative review unifies protein and DNA language models within the AI4Science framework. Beyond standard bioinformatics, biological Large Language Models (bLLMs) fundamentally transform chemical discovery and molecular engineering:

- **Bypassing the Quantum Mechanical Bottleneck:** Traditional computational chemistry relies on expensive Density Functional Theory (DFT) or ab-initio simulations to compute molecular states. bLLMs internalize these biophysical constraints during self-supervised pre-training (3, 4). Transformer-based models have achieved state-of-the-art (SOTA) performance in these tasks due to their self-attention mechanisms and inherent parallelism, which allow them to extract global correlations across long sequences (5-7). By learning the mapping from sequence to physical property, models like ESM-1b and ProtTrans predict molecular forces, secondary structures, and tertiary contacts in milliseconds instead of hours (8-10).
- **Solving the Chemical "Inverse Problem":** Rather than using slow forward simulations to check if a structure works, chemists can leverage bLLMs for De Novo Molecular Design (11,12). Generative decoder-only architectures write entirely novel functional protein and molecular sequences optimized for specific electronic or geometric constraints, bypassing human design bias.
- **Zero-Shot Variant Prioritization:** In data-scarce scenarios like predicting the impact of a rare single-nucleotide polymorphism (SNP) or designing a robust artificial enzyme, bLLMs evaluate mutational costs without task-specific fine-tuning (13-15). By computing embedding-space distances between wild-type and mutated states, chemists can screen thousands of candidates in silico before entering the wet lab.

Review Scope

This paper establishes clear definitions and performance benchmarks for Protein LLMs (ESM-1b, ESMFold, ProteinBERT) and DNA LLMs (DNABERT, Nucleotide Transformer). This study presents a data-driven Comparative Performance Analysis compiled using the empirical datasets and evaluation metrics provided by three authoritative reference studies (13-15), spanning protein foundation models for structure prediction, multi-modal protein models for inverse folding and mutation effect prediction, and DNA foundation models for diverse genomic task.

Finally, we highlight the critical tokenization differences between DNA and protein sequences, evaluate the future potential of ZSL in variant pathogenicity analysis and variant effect prediction, and provide a comprehensive Model Timeline for all bLLMs covered in this study in appendix.

Transformer Architectures in Biological Modeling

The adaptation of the Transformer framework for biological sequence analysis represents a pivotal shift away from purely linguistic processing, focusing instead on capturing the inherent biophysical constraints that govern biological systems. At the core of this architectural adaptation lies the Self-Attention mechanism, which leverages Query (Q), Key (K), and Value (V) vectors to systematically map the internal relationships within biological sequences. Unlike the syntactic parsing used in natural language processing, Multi-Head Attention enables the model to analyze these sequences from multiple distinct biological perspectives concurrently. Specifically, different "heads" within the Multi-Head Attention module can specialize in capturing unique evolutionary and structural signals—including hydrophobicity patterns, secondary structure tendencies, and long-range residue contacts—thereby effectively integrating these diverse biophysical nuances into a unified, cohesive representation (16).

The evolution of Transformer-based biological models has largely centered on the Encoder-only (BERT-style) paradigm, which employs Masked Language Modeling (MLM) as its core self-supervised learning strategy. In contrast to unidirectional decoder architectures, BERT-style models analyze sequences in a bidirectional manner, ensuring that the "context" of any specific amino acid or nucleotide is fully informed by both its upstream and downstream chemical environments. This bidirectional approach allows the model to learn a deep, context-rich "biological grammar," one that is derived from the complex interplay of chemical properties and evolutionary constraints (17). Such bidirectional awareness is critical for enabling foundational models—including ESM-2, ProtBERT, and DNABERT—to extract high-dimensional embeddings that compress the full functional profiles of biological sequences into a single, interpretable vector space. While these Encoder-only models excel at predictive tasks such as variant effect estimation and protein structure prediction (e.g., ESMFold), Decoder-only architectures are specifically prioritized for de novo design applications. These Decoder-only models treat the vast landscape of biological sequence space as a generative domain, "writing" novel functional proteins or synthetic DNA elements by iteratively predicting the next token in a sequence based on the evolutionary grammars learned during pre-training.

Fig. 1.

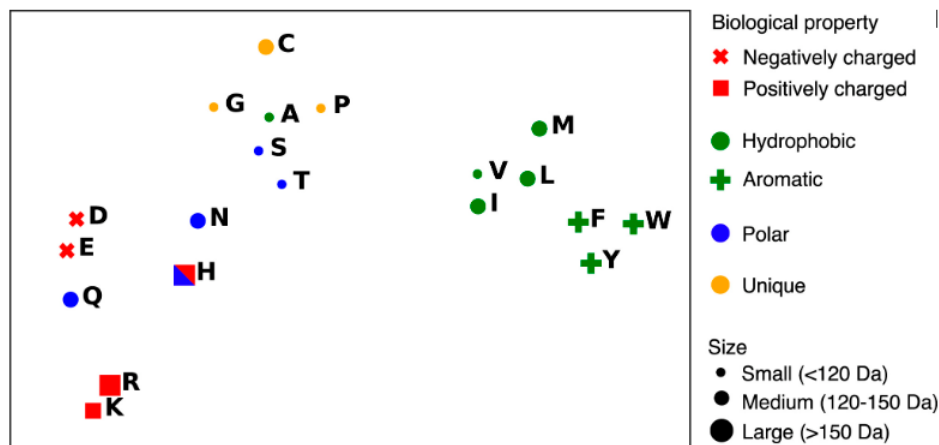


Figure 1. Visualization of amino acid biochemical properties captured by ESM-1b output embeddings. The projection demonstrates that the model autonomously learns to cluster amino acids based on physical and chemical characteristics, including charge (red), hydrophobicity (green), polarity (blue), and molecular size (indicated by marker size) (modified from Ref. (8) Fig. 1).

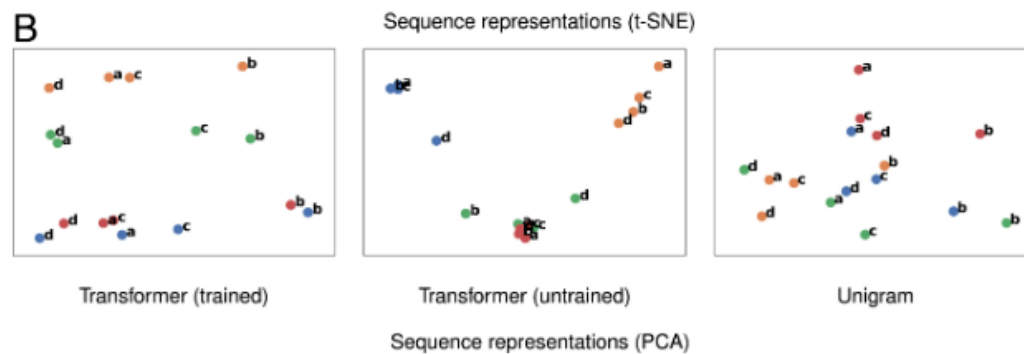


Figure 2. Comparison of protein sequence representation spaces using t-distributed stochastic neighbor embedding (t-SNE) and principal component analysis (PCA). (Left) The trained Transformer (ESM-1b) successfully organizes sequences into a structured coordinate system reflecting biological variations. (Middle) The untrained Transformer and (Right) the Unigram model fail to capture these underlying biological relationships, resulting in disorganized clusters (modified from Ref. (8) Fig. 2B).

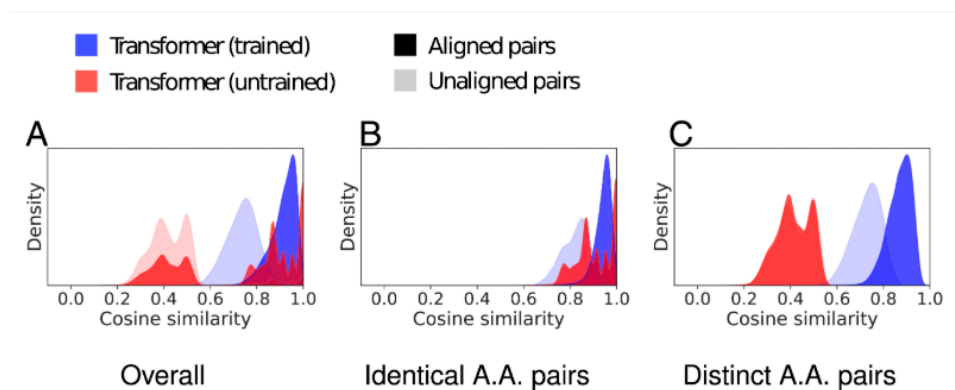


Figure 3. Distribution of cosine similarities for aligned vs. unaligned residue pairs using the ESM-1b Transformer. (A) Overall similarity distribution; (B) similarity for identical amino acid pairs; (C) similarity for distinct amino acid pairs. The trained model (blue) demonstrates a significantly higher density of high-cosine similarity for aligned evolutionary pairs compared to the untrained model (red), indicating successful implicit sequence alignment (modified from Ref. (8) Fig. 3).

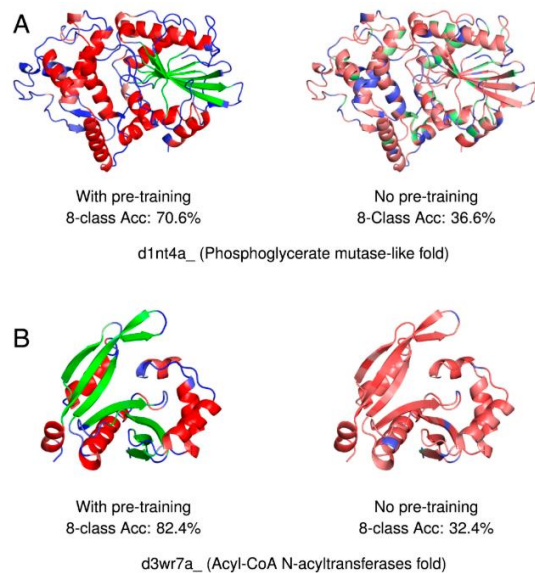


Figure 4. Secondary structure predictions for held-out protein folds A and B. (A) Prediction results for the phosphoglycerate mutase-like fold (d1nt4a_); (B) prediction results for the Acyl-CoA N-acyltransferases fold (d3wr7a_). Comparison between models with pre-training (left) and without pre-training (right) shows that pre-trained embeddings significantly improve 8-class accuracy (Acc), demonstrating that structural signals are linearly encoded within the representation space (modified from Ref. (8) Fig. 4).

Protein language models

ESM-1b laid the foundational groundwork for modern protein language models as a deep Encoder-only Transformer, trained on 250 million diverse protein sequences through unsupervised masked language modeling. The central premise underlying this model is that protein structure and evolutionary processes act as "hidden variables" that determine the patterns observed in amino acid sequences; by learning to predict missing amino acids across a total of 86 billion tokens, the model effectively compresses complex biological rules into its output embeddings—all without the need for explicit 3D structure information. The internal representations generated by ESM-1b reveal a highly organized, multi-scale understanding of biological systems: Figure 1 demonstrates that amino acids naturally cluster together based on their inherent biochemical traits, while Figure 2B uncovers a linearly organized vector space that disentangles the axes of species divergence and orthology. Furthermore, Figure 3 confirms that the model performs implicit sequence alignment, assigning high similarity scores to evolutionarily related residues—even those with different amino acid identities—thereby extracting rich evolutionary signals without relying on computationally expensive Multiple Sequence Alignments (MSAs). To verify that these embeddings contained inherent physical blueprints of proteins, researchers employed linear projections to decode the embeddings, finding that they accurately predict secondary structures and long-range tertiary contacts even for novel, "held-out" protein folds (Figure 4). This breakthrough conclusively proved that high-resolution structural signals are linearly encoded within the language-based representations, laying an essential scientific foundation for subsequent models like ESM-2 and ESMFold to achieve atomic-level 3D protein modeling. Their findings also highlight that larger model sizes and more diverse training datasets enhance overall performance, enabling the generation of novel, biologically active proteins when integrated with advanced gene synthesis technologies (8).

Compared with ESM-1b's 650 million parameters, ESM-2 scales to 15 billion parameters, enabling atomic-resolution structure prediction of biologically active proteins when integrated with gene synthesis technologies. Within the ESMFold architecture, high-dimensional embeddings generated by ESM-2 are processed by the Folding Trunk, a key component that transforms 1D sequence information into dense 2D Pairwise Representations. This Folding Trunk serves as a geometric refiner, constructing complex maps that detail the distances and orientations between every pair of amino acids in the sequence. These spatial insights are then passed to the Structure Module, which leverages the Invariant Point Attention (IPA) mechanism. By treating each amino acid residue as a "rigid body" capable of rotating and moving independently in 3D space, the Structure Module iteratively predicts the exact X, Y, Z coordinates of the protein backbone as well as the specific torsion angles of the amino acid side chains. This streamlined, end-to-end pipeline allowed ESM-2 to achieve evolutionary-scale structural characterization, folding over 617 million proteins in just two weeks and revealing 225 million high-confidence structures that were previously unknown to the scientific community (18).

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procedure FOLDINGBLOCK( $s \in \mathbb{R}^{C_s \times L}$ ,  $z \in \mathbb{R}^{C_z \times L \times L}$ ,  $C_s = 1024$ ,  $C_z = 128$ )
   $b \leftarrow \text{Linear}(z)$ 
   $s \leftarrow s + \text{MultiHeadSelfAttention}(s, \text{bias} = b)$ 
   $s \leftarrow s + \text{MLP}(s)$ 
   $z \leftarrow z + \text{Linear}(\text{Concat}([\text{OuterProduct}(s), \text{OuterDifference}(s)]))$ 
   $z \leftarrow z + \text{TriangularMultiplicativeUpdateOutgoing}(z)$ 
   $z \leftarrow z + \text{TriangularMultiplicativeUpdateIncoming}(z)$ 
   $z \leftarrow z + \text{TriangularSelfAttentionOutgoing}(z)$ 
   $z \leftarrow z + \text{TriangularSelfAttentionIncoming}(z)$ 
   $z \leftarrow z + \text{MLP}(z)$ 
  return  $s, z$ 
end procedure

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Figure 6. Pseudocode for the FoldingBlock mechanism in ESMFold. The procedure details the refinement of single-sequence representations and pairwise representations. Key operations include communication between representations via outer product/difference, triangular multiplicative updates, and triangular self-attention to ensure spatial and geometric consistency within the predicted protein structure (modified from Ref. (18) Supplementary Material Algorithm 1).

The Folding Block functions as an iterative communication hub, synchronizing 1D sequence information with 2D spatial geometry to refine structural predictions. It begins by using the pairwise representation (z) to bias the sequence self-attention (s), enabling the model to focus its attention on residues that are physically close in 3D space. The block then updates the spatial map by projecting sequence features back into the 2D domain through Outer Product and Outer Difference operations. To ensure that the resulting structural blueprints are physically plausible, Triangular Updates and Triangular Attention mechanisms enforce geometric consistency across all residue triplets. This alternating cycle of refinement—synchronizing sequence and spatial information—transforms abstract evolutionary signals embedded in the sequence into high-fidelity 3D structural maps (18).

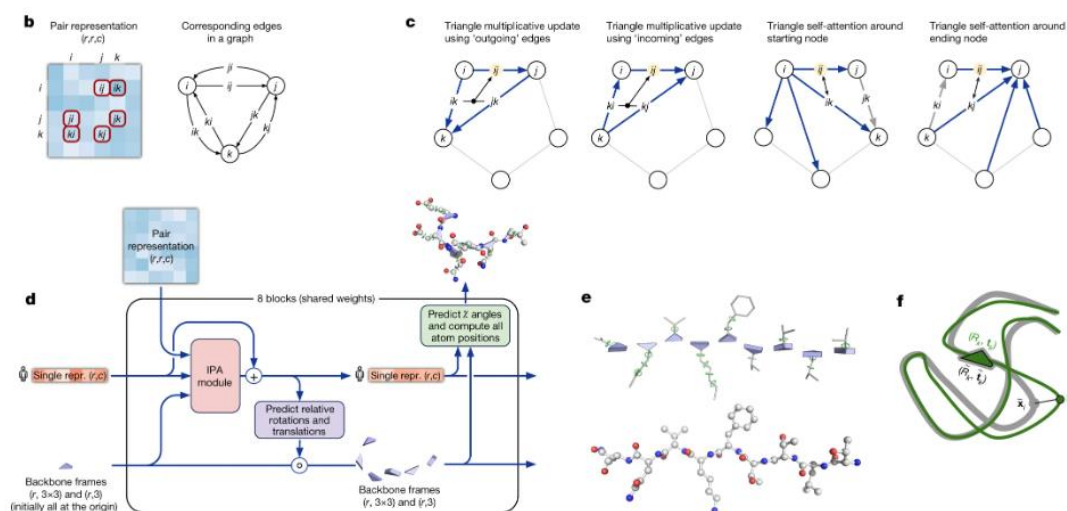


Figure 7. Architecture and operational logic of the Structure Module. (b–c) Illustration of triangle multiplicative updates and triangle self-attention mechanisms applied to the pair representation to maintain geometric consistency through outgoing/incoming edges and nodal attention. (d) The Invariant Point Attention (IPA) module pipeline, showing the iterative refinement of single representations and backbone frames (initially at the origin) to predict relative rotations, translations, and angles θ . (e–f) Transition from localized backbone "gas" frames to a continuous 3D protein fold through supervised coordinate refinement (modified from Ref. (19) Fig. 3b to 3e).

The Structure Module represents the final computational engine responsible for transforming neural representations into tangible 3D protein structures. Operating on a "residue gas" initial state, each amino acid is treated as an independent rigid body defined by rotation and translation frames, rather than being constrained to a strictly linear chain. This approach allows the model to bypass the complex loop-closure problems that often arise during iterative structural refinement. The core of the Structure Module consists of eight distinct blocks that utilize Invariant Point Attention (IPA), which reconciles Single Representations and Pair Representations to update and refine the protein's geometric structure. By performing all calculations in local residue frames, IPA ensures that structural updates remain invariant to global rotation or translation of the entire protein. Following these updates, the module predicts relative rotations and translations to "pose" the backbone frames, eventually calculating the specific angles needed to determine the all-atom positions of the protein. The entire process is optimized using Frame-Aligned Point Error (FAPE) loss, which enforces atomic-level accuracy by aligning the predicted frames with the true experimental frames at the local residue level (19).

While the ESM series excels at capturing the physical and geometric blueprints of proteins, ProteinBERT introduces a specialized architecture that bridges amino acid sequence information with biological function. Recognizing that proteins lack the clear-cut building blocks characteristic of human language, ProteinBERT separates local (residue-level) and global (sequence-level) representations into distinct streams. This dual-stream approach enables the model to handle variable sequence lengths and distant spatial interactions more efficiently than single-stream architectures. A defining innovation of ProteinBERT is its multitasking pretraining strategy: beyond the standard bidirectional language modeling task, the model is pretrained on Gene Ontology (GO) annotation prediction, covering approximately 45,000 terms across three key biological domains—molecular functions, biological processes, and subcellular locations. By integrating manually curated functional labels during the pretraining phase, ProteinBERT learns to directly associate specific sequence patterns with their corresponding biological roles (10).

Logical Transition from Protein LLMs to DNA LLMs

The success of protein language models has demonstrated that sequence-based neural representations can capture intrinsic biological structure and functional properties directly from raw sequence inputs. Inspired by these breakthroughs, researchers extended the Transformer modeling paradigm to genomic datasets, leading to the development of DNA-focused biological foundation models. Although both model families share the underlying Transformer architecture, protein and DNA language models target fundamentally distinct layers of biological organization: protein LLMs interpret molecular folding and biophysical function, while DNA LLMs decipher gene regulatory mechanisms, non-coding biological roles, and long-range chromosomal interactions.

Protein and DNA language models have thus evolved along parallel trajectories, each constrained by the unique linguistic and biochemical properties of their respective molecular sequences. Such fundamental divergence necessitates direct cross-modal comparison to systematically understand the core design principles, inherent limitations, and future research directions of biological foundation models as an integrated research field.

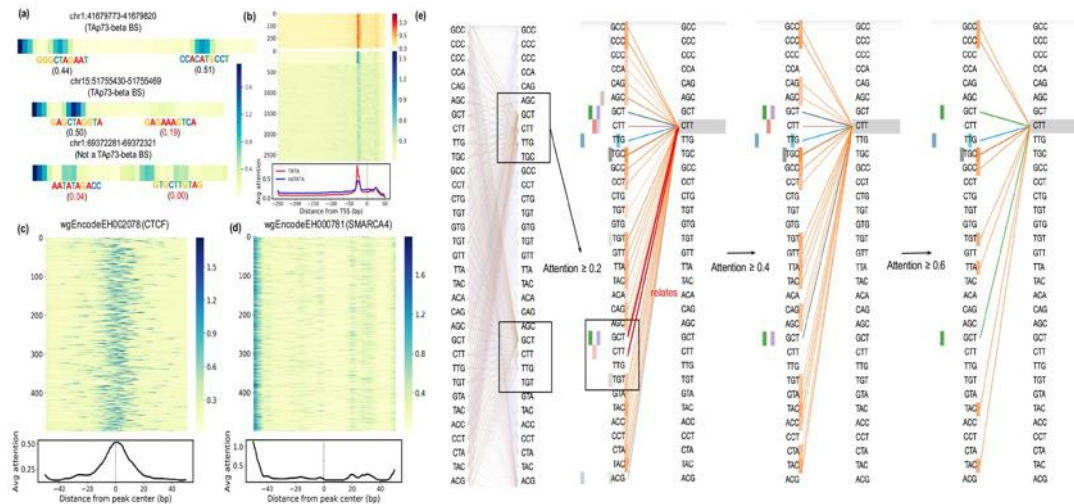


Figure 8. Genome-wide ChIP-seq data analysis and attention mechanism visualization using CombSAFE. (a) Cluster visualization of genomic regions based on features extracted from the Transformer. (b–d) Heatmaps and average signal profiles for transcription factors (e.g., ELF1 and SMARCA4), demonstrating the model's enrichment prediction capability around peak centers. (e) Visualization of attention weights across different thresholds (≥ 0.2 , 0.4, 0.6). The progression illustrates how the model identifies long-range dependencies and core interaction features between functionally similar genomic regions by filtering for the most significant attention links (modified from Ref. (11) Fig. 4).

DNA Language Models

Decoding the inherent regulatory language of non-coding DNA remains one of the central challenges in modern genomics. Complex biological polysemy and long-range contextual dependencies often evade traditional bioinformatic methods, especially under conditions of limited annotated experimental data (11).

DNABERT pioneered pre-trained bidirectional Transformer encoding for global genomic context comprehension by adopting overlapping k-mer tokenization spanning 3 to 6 nucleotides, with the 6-mer configuration recognized as the optimal balance between sequence resolution and computational efficiency. This overlapping tokenization strategy enables transferable learning of both local sequence motifs and distant long-range dependencies, achieving state-of-the-art performance in promoter annotation and transcription factor binding site prediction with only minimal downstream fine-tuning (11).

The auxiliary DNABERT-viz module improves model interpretability by visualizing key genomic features that drive model predictive decisions. As illustrated in Figure 8, the module precisely locates canonical functional elements such as TATA boxes (8b) and transcription factor binding loci (8a), where elevated attention weights closely align with experimentally validated genomic positions. Beyond feature annotation, DNABERT-viz also functions as a ChIP-seq quality control tool, differentiating high-quality datasets with concentrated, centralized attention patterns (8c) from low-quality datasets exhibiting scattered, dispersed attention distributions (8d). Multi-head attention visualizations (8e) further uncover the implicit grammatical rules of DNA sequences by displaying long-range contextual relationships and token convergence at functional dimeric binding sites (11).

Even so, DNABERT carries notable inherent limitations. Sanabria et al. (2024) demonstrated that while DNABERT effectively learns individual token-level sequence identity, its performance degrades when processing extended genomic contexts due to the constraints of overlapping k-mer tokenization, which restricts its applicability to tasks requiring deep long-range contextual comprehension (20).

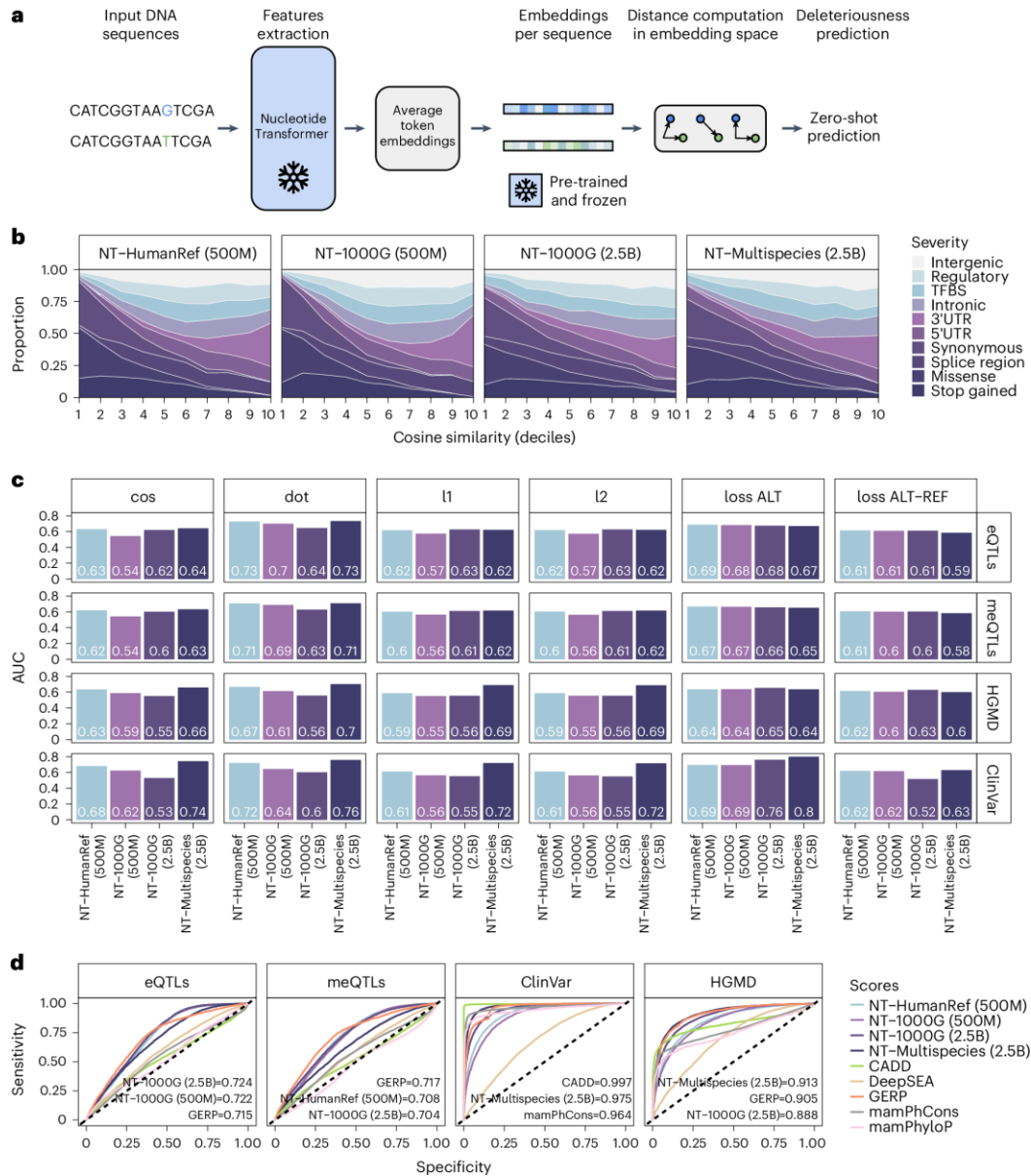


Figure 9. Zero-shot prediction of genetic variant deleterious effects using the Nucleotide Transformer (NT). (a) Schematic of the inference pipeline: input DNA sequences are processed through a frozen pre-trained model to extract averaged token embeddings, followed by distance computation in the embedding space for zero-shot prediction. (b) Distribution of variant severity types (e.g., missense, stop gained, intronic) across cosine similarity deciles for different NT model variants. (c) Performance comparison (AUC) of various scoring metrics (Cosine, Dot, L1, L2) across eQTL, meQTL, HGMD, and ClinVar benchmarks. (d) Receiver operating characteristic (ROC) curves demonstrating that the NT-Multispecies (2.5B) model achieves superior sensitivity and specificity compared to traditional supervised scores such as CADD, GERP, and DeepSEA (modified from Ref. (12) Fig. 4).

The Nucleotide Transformer (NT) marks a substantial advancement in genomic foundation modeling, constructed with 2.5 billion parameters and trained on 3,202 human genomes plus 850 phylogenetically diverse species genomes. By integrating both intra-species genetic polymorphism and inter-species evolutionary variation, NT captures functionally critical genomic signatures that exhibit strong generalization across human genomic prediction tasks, even in low-label data scenarios (12).

This embedded evolutionary “biological intuition” enables powerful zero-shot genetic variant prioritization. By calculating embedding-space distance metrics between wild-type and mutated DNA sequences, the model identifies functional and regulatory disruptions without task-specific supervised fine-tuning. Figure 19b experimentally validates this zero-shot framework: variants with the lowest cosine similarity scores correspond to the most severe mutational impacts (e.g., stop-gained and missense mutations), whereas intergenic variants with minimal biological impact maintain highly stable embedding similarity (12).

Table 1: Comparative Analysis of Tokenization Strategies, Semantic Density, and Computational Architectures in Protein vs. DNA Language Models.

Feature	Protein Language Models (pLMs)	DNA/Genomic Language Models (dLMs)
Primary Token Unit	Single Amino Acid (AA)	K-mers, BPE, or Multi-nucleotide units
Semantic Density	High: 20 residues with distinct biochemical properties (e.g., charge, hydrophobicity).	Low: 4 nucleotides (A, C, G, T) requiring clustering for regulatory meaning.
Tokenization Strategy	Direct Mapping: Natural 1-to-1 alignment between residues and tokens.	Aggregation: Overlapping k-mers (e.g., 6-mer) or Byte Pair Encoding (BPE).
Information Resolution	High: Pinpoints single-residue mutations (SNPs) with atomic precision.	Lower/Blurred: Mutations are "smeared" across multiple overlapping k-mer tokens.
Long-Range Source	3D Folding: Distant residues contacting in physical 3D space.	Distal Regulation: Enhancers/promoters interacting across 10kb to 1Mb.
Computational Priority	Vertical Depth: Atomic-level geometric refinement (e.g., ESMFold).	Horizontal Span: Capturing massive genomic contexts via efficient attention.
Effective Context	Relatively Short (< 2k tokens) sufficient for most structural domains.	Extremely Long (> 10k to 1M tokens) required for 3D chromatin organization.
Key Bottleneck	Modeling high-resolution biophysical constraints.	Token Redundancy and quadratic scaling $O(N^2)$ complexity.

Tokenization Differences Between Protein and DNA LLMs

The tokenization challenge constitutes a critical point of divergence between protein and DNA language models, directly dictating how each architectural framework allocates computational resources. Protein sequences naturally lend themselves to amino acid-level tokenization, where each of the 20 standard amino acid residues functions as a high-density semantic unit. This direct one-to-one mapping enables models such as ESM-1b to internalize complex structural and evolutionary constraints without the need for extensive preprocessing (8). Given that protein functional domains are relatively compact in size, pLMs reallocate their computational budgets away from sequence length accommodation and toward vertical depth—focusing on deeper feature extraction and atomic-level geometric precision in structural predictions.

In contrast, DNA language models (dLMs) grapple with a fundamental "low information density" challenge, stemming from the limited four-nucleotide alphabet (A, C, G, T). To extract meaningful biological signals from this sparse vocabulary, dLMs employ specialized tokenization strategies such as overlapping k-mers or Byte Pair Encoding (BPE). While DNABERT's 6-mer tokenization approach effectively captures local sequence motifs, it introduces inherent "token redundancy": a single nucleotide mutation becomes blurred across six consecutive k-mer tokens, thereby obscuring the precise causal signal of the mutation (20-21). Furthermore, because critical genomic interactions—such as enhancer-promoter regulatory dynamics—occur across vast linear genomic distances exceeding 10kb, dLMs must prioritize horizontal span (i.e., context expansion) over the depth of individual token representation (22).

This inherent difference creates a fundamental architectural trade-off: pLMs prioritize high-resolution biophysical depth to accurately predict protein folding, while dLMs sacrifice local sequence resolution in favor of global "reach" to capture 3D genomic organization. Recent advancements in sub-quadratic architectures (e.g., HyenaDNA) seek to break this bottleneck by enabling single-nucleotide resolution across megabase-scale genomic contexts. By moving beyond fixed k-mer tokenization, these innovative models aim to reconcile the need for local base-pair precision with the requirement to capture long-range dependencies inherent to the "global genomic grammar."

The Necessity of Zero-Shot Learning in Addressing Biological Data Scarcity

Zero-shot learning (ZSL) represents a transformative capability in biological language models, empowering these systems to perform complex predictive and analytical tasks without task-specific training or labeled data. By leveraging extensive evolutionary patterns and core biological principles internalized during pre-training, these models can generalize their learned knowledge to unseen biological scenarios through their high-dimensional internal representations (23).

The fundamental importance of ZSL stems from the pervasive challenge of data scarcity in biological research; traditional supervised learning approaches require vast quantities of annotated data that are often prohibitively expensive to generate or entirely impossible to obtain—particularly for emerging pathogens or rare biological phenomena. ZSL circumvents this critical bottleneck by making predictions based on fundamental biological "grammar" rather than relying on specific labeled training examples (24). Furthermore, it enhances computational efficiency and scalability by eliminating the need for repetitive, task-specific fine-tuning across distinct biological tasks (25).

In the domain of variant pathogenicity analysis, ZSL is particularly powerful for evaluating genomic-level DNA disruptions. Rather than relying on historical clinical labels or curated mutation datasets, models such as the Nucleotide Transformer v2 (NT v2) and DNABERT-2 assess the functional impact of a mutation by calculating the "evolutionary cost" or embedding-space distance metrics between wild-type and mutated DNA sequences (26-27). This approach enables the immediate identification of deleterious disruptions in non-coding genomic regions that have never been experimentally characterized or annotated. Similarly, variant effect prediction (VEP) focuses on the proteomic level, aiming to determine how amino acid changes alter protein stability and functional activity. Models like ESM-1v and ESM-2 provide a crucial advantage in time-sensitive applications—such as responding to emerging disease outbreaks—by rapidly assessing the functional impacts of new viral variants without the need to wait for specialized training data (18, 28). The integration of these ZSL-capable models has a significant impact on precision medicine, especially in the context of rare diseases where experimental data is extremely limited. By utilizing unsupervised architectures or generative models such as EVE, researchers can prioritize potentially pathogenic genetic variants even when only a handful of clinical cases exist worldwide (13). This ability to identify pathogenic culprits within a "haystack" of neutral mutations dramatically accelerates clinical diagnostics and the discovery of targeted therapeutic strategies. Consequently, ZSL transforms biological language models from static analytical tools into dynamic systems capable of real-time biological inference across uncharted research territories.

Table 2 Taxonomy of Downstream Applications for pLMs and dLMs, Highlighting the Transition from Sequence Analysis to Functional, Structural, and Generative Biological Engineering

Application Domain	Protein Language Models (pLMs)	DNA/Genomic Language Models (dLMs)
Structural Inference	3D Structure Prediction: Atomic-level folding without MSAs (ESMFold, Chai-1).	3D Genome Organization: Identifying TADs and chromatin folding (Caduceus-Ph, HyenaDNA).
Functional Mapping	Functional Classification: Categorizing enzymes, antibodies, and hierarchies (ProteinBERT).	Regulatory Elements: Pinpointing promoters and enhancers (DNABERT-2, NT-v2).
Variant Analysis	Pathogenicity & Stability: Analyzing SAVs, nonsense, and frameshift mutations (ESM-1v, TransPPMP).	Variant Prioritization: Evaluating mutation deleteriousness in coding/non-coding regions (NT-v2).
Clinical & Biomedical	Immunology: B-cell epitope mapping for vaccine design (BepiPred-3.0).	Precision Medicine: Identifying therapeutic targets in complex pathologies like cancer (NT-v2).
Molecular Interactions	Target-Driven Design: Engineering high-affinity binders and ligands (AlphaProteo, Chai-1).	Expression Regulation: Predicting transcription rates and cell-type activity (GROVER).
Generative Design	De Novo Protein Engineering: Creating novel functional sequences (ProGen, ProtGPT2).	Synthetic Genomics: Engineering synthetic regulatory elements (DNAGPT).

Downstream Applications of Protein Language Models

The downstream applications of protein language models (pLMs) have redefined the boundaries of computational proteomics, advancing from basic sequence analysis to high-fidelity structural and functional inference. ESMFold exemplifies this transformative shift by leveraging ESM-2 embeddings to achieve atomic-level 3D protein structure prediction, effectively bypassing the need for computationally intensive multiple sequence alignments (MSAs) (18).

In contrast, traditional approaches like AlphaFold2 demonstrate groundbreaking success in protein structure prediction by leveraging multiple sequence alignments (MSAs) to infer evolutionary relationships, significantly improving accuracy. However, its reliance on high-quality MSAs limits performance in cases with sparse sequence data, highlighting the need for advancements in low-MSA or MSA-free methods (19). Similarly, OmegaFold addresses the limitations of MSA-based protein structure prediction for orphan and fast-evolving proteins by leveraging a protein language model and geometry-inspired transformer, achieving high-resolution predictions from single sequences comparable to AlphaFold2 (29).

Beyond structural inference, models such as ProteinBERT enable precise functional classification, allowing researchers to categorize proteins into specific biological roles—such as enzymes or antibodies—based on learned evolutionary and functional hierarchies (10). Furthermore, specialized models including BepiPred-3.0 and TransPPMP demonstrate the broad versatility of ESM-derived embeddings: the former enhances vaccine design through precise B-cell epitope mapping, while the latter provides critical clinical diagnostic support by analyzing the pathogenicity of complex nonsense and frameshift mutations (28, 30).

Additionally, variant effect prediction has emerged as a cornerstone of pLM utility, with specialized models like ESM-1v performing zero-shot assessments of how single amino acid substitutions influence protein stability or lead to functional loss—providing critical insights into disease mechanisms at the molecular level (31).

These capabilities also facilitate the exploration of uncharted biological sequence space through de novo protein design. Generative models such as ProGen and ProtGPT2 leverage learned evolutionary grammars to engineer novel, functionally active protein sequences (32-33). More advanced multimodal approaches like ESM3 demonstrate the ability of multimodal protein language models to generate functional proteins far from known sequences, exemplified by the successful creation of a highly fluorescent protein with substantial evolutionary divergence (34). Similarly, DPLM-2 addresses the limitations of separate multimodal protein modeling methods by jointly learning sequence-structure relationships through a single framework, introducing a novel tokenizer for structural learning and leveraging a warm-up strategy for effective use of evolutionary and structural data (35).

Moving toward target-specific therapeutic development, newer frameworks like AlphaProteo represent a key transition toward target-driven design, engineering high-affinity protein binders tailored to the 3D surfaces of disease-related receptors (36). Finally, emerging multi-modal models like Chai-1 further enhance this ecosystem by predicting complex protein-ligand interactions without relying on MSAs, effectively integrating pLMs into high-throughput drug discovery pipelines (37).

Downstream Applications of DNA Language Models

Parallel advancements in DNA language models (dLMs) have transformed genomic research, shifting it from simple sequence mapping to complex functional and clinical inference. The early foundational work established by DNABERT-2 and the Nucleotide Transformer (NT-v2) is frequently deployed for promoter and enhancer identification, pinpointing the precise "on/off switches" that govern gene transcription (26,27).

By analyzing the contextual grammar embedded within non-coding sequences, these dLMs identify distal regulatory elements that modulate gene expression across vast genomic distances—all without requiring extensive experimental annotations.

The clinical utility of dLMs has expanded into variant effect quantification, where models evaluate mutation deleteriousness through high-dimensional embedding representations. NT-v2 enables zero-shot prioritization of genetic variants, correlating embedding distances with the severity of functional disruption in both coding and non-coding genomic regions—a critical capability for identifying therapeutic targets in complex pathologies such as cancer.

Beyond point mutations, architectures including Caduceus-Ph and HyenaDNA utilize sub-quadratic scaling to capture long-range genomic dependencies, excelling in Topologically Associating Domain (TAD) recognition (38-39).

These models offer valuable insights into the genome's 3D spatial organization, revealing how distant DNA segments physically interact to regulate gene expression.

Furthermore, GROVER has pushed the boundaries of gene expression prediction by treating DNA as a language with a learned vocabulary of multi-nucleotide tokens. By capturing the genome's statistical "grammar," GROVER predicts transcription rates across diverse cell types, elucidating how sequence context tunes gene activity (40). These insights facilitate de novo genomic design via generative frameworks like DNAGPT, enabling the engineering of synthetic regulatory elements with precise expression profiles (41). By bridging the gap between raw genomic data and clinical medical interventions—from gene annotation to pathogenicity analysis—dLMs provide a systematic framework for accelerating precision therapies and unraveling the complexities of the "dark genome."

Comparative Performance Analysis of Protein Foundation Models in Structure Prediction Tasks

A comprehensive benchmarking study involving 1,337 protein chains deposited in the Protein Data Bank (PDB) between 2022 and 2024 evaluated the predictive capabilities of three leading models: AlphaFold2, ESMFold, and OmegaFold. The evaluation employed key performance metrics, including the Root Mean Square Deviation (RMSD)—which measures the average physical distance between corresponding atoms, with lower values indicating higher structural precision—and the Template Modeling score (TM-score)—which quantifies global topological similarity on a scale of 0 to 1, where higher scores reflect greater predictive accuracy. Additionally, the study utilized the Global Distance Test–Total Score (GDT-TS), a metric widely employed in the Critical Assessment of Structure Prediction (CASP) to assess the alignment between predicted and experimentally determined protein structures (14).

The findings demonstrated that AlphaFold2 achieved superior predictive accuracy, recording a median TM-score of 0.96 and a median RMSD of 1.30 Å, thereby outperforming both ESMFold (1.74 Å) and OmegaFold (1.98 Å). The analysis further revealed that prediction precision is highly sensitive to the experimental methodology used for structural determination. While all three models performed optimally on proteins characterized via X-ray crystallography, structures determined through nuclear magnetic resonance (NMR)—which are typically associated with shorter amino acid sequences—presented the most significant challenges to predictive performance (14).

Notably, the models displayed distinct sensitivities to evolutionary data availability; proteins lacking specific family annotations significantly hindered AlphaFold2's efficacy, as the model relies heavily on Multiple Sequence Alignments (MSAs) and established structural templates. In contrast, the performance of protein language models (pLMs) such as ESMFold and OmegaFold remained largely consistent in these scenarios, as they leverage a learned "protein grammar" rather than conducting an evolutionary sequence search. This fundamental architectural distinction enables alignment-free methods to maintain high predictive stability when forecasting the structures of de novo designed proteins or those without recognized homologous sequences (14).

Comparative Performance Analysis of Multi-Modal Protein Foundation Models: Structure Prediction, Inverse Folding & Mutation Effect Prediction

ProteinReasoner represents a significant paradigm shift in multi-modal protein language modeling through the introduction of an explicit Chain-of-Thought (CoT) reasoning framework (15). Unlike ESM3, which utilizes an end-to-end learning approach (34), or DPLM-2, which focuses primarily on the integration of sequence and structural data (35), ProteinReasoner synthesizes three distinct biological modalities: amino acid sequences, 3D structural conformations, and evolutionary profiles. By utilizing the evolutionary profile as an intermediate reasoning step, the model simulates inherent biological logic instead of performing simple data mapping, resulting in superior performance across both protein understanding and de novo design tasks (15).

To evaluate these multi-modal capabilities, a comparative study assessed ProteinReasoner, ESM3, and DPLM-2 across four standardized benchmarks: CAMEO, CASP14, CASP15, and a PDB date-based holdout split. In zero-shot structure prediction, performance was quantified using RMSD and TM-score—consistent with established structural prediction metrics. Despite its smaller parameter count, ProteinReasoner-650M in externally guided mode consistently outperformed all baseline models. When compared to DPLM-2-650M, it achieved a TM-score up to 26.3% higher on CASP14 and 14.2% higher on CAMEO, while improving RMSD by up to 0.81 Å. Against the larger ESM3-Open-1.4B, ProteinReasoner-650M recorded a 10.3% higher TM-score on CASP14 (15).

In the context of zero-shot inverse protein folding—where the goal is to design a sequence that folds into a target 3D structure—the models' capacity for sequence design was measured using two key metrics: Amino Acid Recovery (AAR), which tracks the accuracy of natural sequence recovery at each residue position, and the self-consistency TM-score (scTM), which assesses structural success by comparing the original input structure to one predicted from the newly generated sequence via AlphaFold2. ProteinReasoner-650M demonstrated exceptional performance in structural consistency, achieving scTM scores of 0.786 on the combined CAMEO+CASP dataset and 0.918 on the PDB split. Notably, against DPLM-2-3B, ProteinReasoner-650M achieved a 1.9% higher scTM despite a 5.1% lower AAR, suggesting a superior ability to identify solutions that are structurally compatible yet sequence-diverse (15).

For mutation effect prediction via the ProteinGym benchmark, the models predicted the functional consequences of genetic variations. Performance was evaluated using Spearman correlation to measure rank-order agreement between predicted and observed fitness scores, alongside the Area Under the Receiver Operating Characteristic curve (AUROC) to determine the model's ability to differentiate beneficial mutations from deleterious ones. ProteinReasoner-650M outperformed ESM3-Open-1.4B across these metrics, proving that its reasoning-driven multi-modal integration is highly efficient for the comprehensive tasks of protein understanding and design (15).

Comparative Performance Analysis of DNA Foundation Models

A comprehensive evaluation of five state-of-the-art DNA foundation models—DNABERT-2, NT-v2, HyenaDNA, Caduceus-Ph, and GROVER—was conducted using a standardized zero-shot approach across 57 sequence classification datasets. These datasets represented diverse genomic categories, including human and multi-species genome sequences as well as single-species and multi-species epigenetic traits. The findings revealed that architectural specializations significantly influence model performance across distinct biological tasks. For transcription factor binding site (TFBS) prediction—a critical task for identifying disease-associated regulatory disruptions—Caduceus-Ph emerged as the superior model, achieving the highest Area Under the Receiver Operating Characteristic (AUC) scores. AUC (Area Under the Curve) measures the ability of a binary classification model to distinguish between two classes (e.g., binding vs. non-binding sites). In contrast, epigenetic modification detection—which identifies chemical markers like 5-methylcytosine and N6-methyladenosine, essential for cancer diagnosis and developmental biology—exhibited species-dependent performance results. NT-v2 demonstrated exceptional performance across multiple species, whereas DNABERT-2 proved most effective for yeast-specific epigenetic mark prediction (13).

Further task-specific analysis highlighted the strengths of discrete model architectures in identifying structural and clinical genomic features. DNABERT-2 achieved superior performance in splice site prediction, underscoring its proficiency in recognizing the exon-intron boundaries that govern mRNA processing; such accuracy is vital for designing gene therapies for diseases caused by splicing defects. For clinical applications, NT-v2 established itself as the leading architecture for pathogenic variant identification, attaining an AUC of 0.73 in distinguishing disease-causing mutations from benign variants—a capability that serves as a cornerstone of precision medicine. Additionally, while most models focused on binary classification tasks, HyenaDNA showed remarkable capabilities in multi-class regulatory region distinction, facilitating the simultaneous identification of diverse regulatory elements (e.g., promoters, enhancers). Collectively, these results suggest that the selection of a DNA foundation model must be precisely aligned with the functional requirements of the downstream genomic or genetic task (13).

Limitation

The biological LLM landscape evolves at an unprecedented pace, making comprehensive coverage challenging. Some emerging models and specialized applications remain outside this review's scope due to the field's rapid advancement (42). Current performance evaluations rely primarily on existing benchmarks, which suffer from data contamination, cultural biases, and inadequate assessment of dynamic biological environments (43). These limitations necessitate transitioning to broader, multi-dimensional evaluation frameworks that can accurately assess model performance under diverse biological conditions. Novel domain-specific benchmarks are emerging to address these gaps, offering more robust methodologies for specialized applications (44).

Conclusion

The evolution of biological Large Language Models represents a significant advancement in interpreting complex biological sequences, recognizing proteins and DNA as interrelated molecular languages. This review highlights the critical importance of tailored tokenization strategies and zero-shot learning capabilities for capturing long-range dependencies and functional insights within biological data. In data-scarce scenarios, these models prove invaluable for variant pathogenicity analysis and variant effect prediction. The integration of evolutionary-informed sequence learning has enabled remarkable progress in protein design, variant prediction, and structure determination, signaling substantial potential for precision medicine applications. However, challenges in model interpretability, clinical validation, and translating computational predictions to real-world implementations remain key hurdles requiring continued research.

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Appendix

Combined Protein Language Models Master Table (2019 – 2024)

Model Name (Year)	Architecture Paradigm	Tokenization Strategy	Core Technical Innovation	Key Benchmarks & Roles
ESM-1b (2019)	Encoder-only	Amino acid-level residue tokenization	Bidirectional Masked Language Modeling (MLM).	Proved structure and evolution act as hidden variables in sequences.
ProtTrans / ProtBERT (2021)	Encoder-only (BERT-based)	Amino acid-level residue tokenization	High-dimensional embedding space compression.	Laid foundational benchmarks for high- density feature extraction.
ESM-1V (2021)	Encoder-only (Variant- optimized)	Amino acid-level residue tokenization	Independent likelihood calculation for variant tracking.	High-performance Zero-shot Variant Effect Prediction (VEP).
AlphaFold (2021)	Evoformer Architecture	Amino acid-level residue tokenization	Explicit constraints using evolutionary profiles via MSAs.	Gold standard for structure prediction (Median TM-score: 0.96, RMSD: 1.30 Å on CASP14 targets).
ProteinBERT (2022)	Dual-stream (Local/Global)	Amino acid-level residue tokenization	Pre-trained simultaneously with 45,000 Gene Ontology (GO) terms.	Bridges the sequence-to-function gap for macro functional categories.
ESM-2 / ESMFold (2022)	Encoder-only (15B params)	Amino acid-level residue tokenization	Geometric refinement via Folding Trunk & Invariant Point Attention (IPA).	High-throughput structural pipeline; folded over 600M proteins in 2 weeks.
OmegaFold (2022)	Single- sequence Transformer	Amino acid-level residue tokenization	Single-sequence structural inference network.	Bypasses MSAs entirely; highly robust for orphan proteins or design.
EVE (2023)	Variational Autoencoder (VAE)	Amino acid-level residue tokenization	Evolutionary encoding and coupling analysis.	Prioritizes pathogenic variants in rare diseases with sparse clinical data.
BepiPred-3.0 (2024)	Specialized Epitope Predictor	Amino acid-level residue tokenization	Deep learning-based antibody-antigen mapping leveraging ESM-2 embeddings.	Specialized mapping of linear and conformational B-cell epitopes for vaccine design.
ESM3 / ESM3- Open (2024)	Multi-modal Unified Tokenization	Sequence, Structure, & Function Tokenizer	Simulates parallel tracks of life's properties (sequence, structure, function).	Generated novel functional Green Fluorescent Proteins (GFP).
DPLM-2 (2024)	Discrete Diffusion Framework	Lookup-free Quantization Tokenizer	Simultaneous co- generation of sequence and 3D space coordinates.	Massive 3B parameters; achieves high Amino Acid Recovery (AAR) rate.
ProteinReasoner (2024)	Chain-of- Thought (CoT) Multi-modal	Sequence, Structure & Evolutionary Tokenizer	Integrates evolutionary profile as an intermediate reasoning step to simulate biological logic.	Outperformed DPLM- 2 by 26.3% in TM- score on CASP14 zero-shot tasks; superior on ProteinGym.

AlphaProteo (2024)	Generative Protein Engineering System	Amino acid tokenization	High-affinity structural binder matching and targeted de novo generation.	Engineers high-affinity binders for specific target receptors.
Chai-1 (2024)	Multi-modal Interaction Pipeline	Structural/Molecular Tokenizer	Integrates physical structural tokens with raw text tokens (chemical ligands).	Predicts protein-multimer and protein-ligand interactions directly for drug discovery.

Transposed DNA LLMs Master Table

Model Name (Year)	Architecture Paradigm	Tokenization Strategy	Core Technical Innovation	Key Benchmarks & Roles
DNABERT (2021)	Encoder-only Transformer	Overlapping k -mer tokenization	Attention weight visualization module (DNABERT-viz).	Pioneered non-coding DNA decoding; targets promoters and TFBS.
DNABERT-2 (2023)	Upgraded Encoder-only	Optimized nucleotide-level tokenization	Mitigates token redundancy to clear causal signal blur.	Superior accuracy in splice site prediction; best for yeast epigenetics.
Nucleotide Transformer (NT / NT-V2) (2023)	Massive Encoder-only (2.5B params)	Fixed k -mer multi-species tokenization	Scaled across thousands of multi-species genomes.	Achieved AUC of 0.73 in clinical pathogenic variant identification.
HyenaDNA (2023)	Implicit Convolutional Network (CNN)	Single-nucleotide resolution	Replaces self-attention to unlock sub-quadratic scaling.	Achieved megabase-scale context windows; best for global regulation.
GROVER (2023)	Statistical Grammar Learner	Genome dictionary tokenization	Captures statistical properties of DNA texts.	Learns genomic statistical rules to predict transcription rates.
DNAGPT (2023)	Decoder-only Generative	Generative sequence tokenization	Treats sequence space as an open generative domain.	Enables de novo design of synthetic regulatory elements.
Caduceus / Caduceus-Ph (2024)	Bidirectional Mamba / State Space Model (SSM)	Single-nucleotide resolution	First bidirectional SSM architecture for genomics.	Highest AUC scores for TFBS prediction; maps Topologically Associating Domains (TADs).